

Problem E. Equação

Time Limit 3000 ms

Mem Limit 262144 kB

Given an integer x . Your task is to find out how many positive integers n ($1 \leq n \leq x$) satisfy

$$n \cdot a^n \equiv b \pmod{p},$$

where a, b, p are all known constants.

Input

The only line contains four integers a, b, p, x ($2 \leq p \leq 10^6 + 3, 1 \leq a, b < p, 1 \leq x \leq 10^{12}$). It is guaranteed that p is a prime.

Output

Print a single integer: the number of possible answers n .

Examples

Input	Output
2 3 5 8	2
Input	Output
4 6 7 13	1
Input	Output
233 233 10007 1	1

Note

In the first sample, we can see that $n = 2$ and $n = 8$ are possible answers.